The Dual-Band SAR Image Fusion-Based Foliage-Penetrating Target Detection Method

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***Abstract*— The low-band synthetic aperture radar (SAR) is a system that can present foliage-penetrating (FP) targets and exposed strong-scattering targets while the high-band SAR system can present exposed targets and texture information except for FP targets. Therefore, the different features of co-registered low-band and high-band SAR images can theoretically reveal FP targets with some disturbances. However, the co-registered dual-band SAR images cannot do the difference operator directly for the distinct statistical property of the SAR image pair. The traditional method is to transfer the background texture from the high-band SAR image to the low-band SAR image by the cycle-consistent generative adversarial network (CGAN) method. However, CGAN would generate discontinuity and mistakes in transferring the large-scale texture information, and the complex network would heavily increase the time complexity of the application. To overcome this issue, we first innovatively introduce a fast and accurate texture-transferring method based on the dual-band SAR image fusion (DBIF), then we further combine the DBIF-based texture-transferring method with the difference operator and the threshold segmentation, and finally we get a novel DBIF-based FP target detection (DBIFFPTD) method. To verify the feasibility of the DBIF-based texture transfer method, we discuss the reflection form of different landscape objects at dual-band waves and estimate the theoretical texture transferring the image. Experiments on the open dual-band AIR-MD-SAR dataset and the independently measured dual-band SAR dataset show that the novel DBIF performs better than CGAN in transferring texture information of dual-band SAR images. Besides, dual-band linear SAR (LSAR) and circular SAR (CSAR) FP experiments are both conducted to show that the proposed DBIFFPTD has wide applicability in the flight track and is superior to the traditional low-band double-parameter constant false alarm rate (DCFAR) detector-based FPTD (DCFARFPTD) method and the dual-band CGAN-based FPTD (CGANFPTD) method.**

***Index Terms*— Cycle-consistent generative adversarial network (CGAN), difference operator, foliage-penetrating (FP) targets, image fusion, threshold segmentation, transfer.**

I. Introduction

SYNTHETIC aperture radar (SAR) systems have been widely utilized for electronic surveillance, emergency rescue, and all-weather remote sensing. Most SAR images are taken from a single-path, single-channel, single-frequency, and single-polarized system for simple operation and cheap manufacture. However, the performance of an automatic target recognition system could be enhanced by the additional information of targets which is commonly achieved by the multipass [1] collection system from different aspect angles, the single-pass collection system with the multiband mode [2], and the single-pass collection system with the multipolarization mode [3].

For the foliage penetration of the low-band (P-band or L-band) radar signal, the low-band SAR system can image foliage-penetrating (FP) targets. To further detect targets hidden by the foliage, the double-parameter constant false alarm rate (DP-CFAR) detector [4] was taken to detect FP targets from clutters. However, strong-scatter exposed targets in low-band SAR images would disturb the detection result of FP targets, thus it is difficult to directly detect FP targets with a low false alarm rate from a single low-band SAR image. To alleviate the false alarm rate of the detection result, the high-band (Ku-band or Ka-band) SAR system with a high-resolution imaging ability was introduced to obtain a reference SAR image [5]. For the high-band SAR signal lacks the FP ability, the high-band SAR image mainly presents exposed targets and texture information. Therefore, conducting the difference operator [6] to the low-band and high-band SAR images can eliminate false alarms generated by strong-scatter exposed targets.

However, dual-band SAR images cannot directly make a difference in detecting FP targets due to the distinct statistical properties and texture characteristics. To address this issue, the cycle-consistent generative adversarial network (CGAN) [7] utilized in the multimodal change detection was conducted to learn the texture transferring relation between the dual-band SAR images [5]. Despite the texture information of the high-band SAR image can be transferred to the low-band SAR image by CGAN, the fake high-band SAR image with the transferred texture would generate some discontinuities for the large-scale SAR image because the weight coefficient of CGAN trained by image patches which are split from the large-scale image would retain the splicing trace in the fake high-band SAR image. Besides, the training of CGAN depends on a heavy computational complexity which would limit the development of CGAN in the FP target detection (FPTD).

To address the above issues, inspired by the dual-band SAR image fusion method (DBIF) [8], which considers overall texture information and target information in the large-scale

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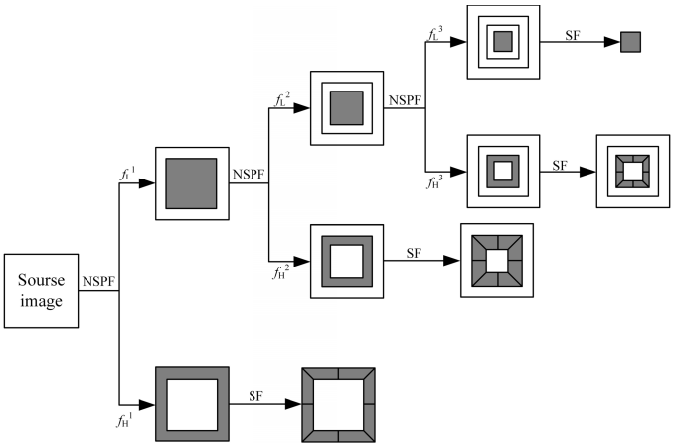


Fig. 1. Flowchart of three-level NSST.

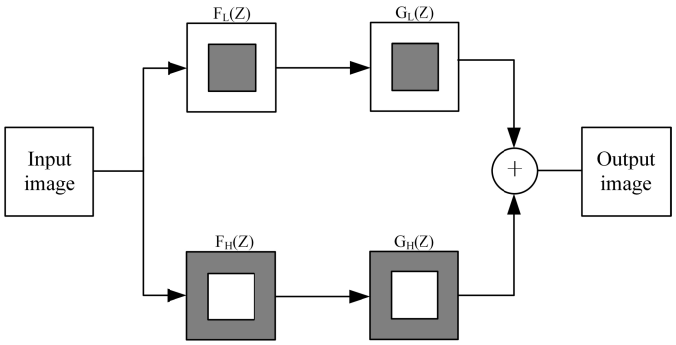


Fig. 2. Diagram of NSPF.

scene, we adopt a novel texture transferring method that fuses dual-band SAR images to generate a fake high-band SAR image. The proposed FPTD method first makes the low-band SAR image obtain the fake texture information which is similar to the texture detail of the high-band SAR image by the dual-band image fusion, then measures the difference feature of the high-band SAR image and the fake high-band SAR image, and finally extracts FP targets by the threshold segmentation.

Applying DBIF, difference operator, and threshold segmentation hierarchically to dual-band SAR images, generating a novel DBIF-based FPTD (DBIFFPTD) method. The main contributions of this article are as follows.

1) We propose a novel DBIF-based texture-transferring method to realize an efficient texture transfer between dual-band SAR images.
2) We combine the proposed texture transferring method with the change detection to realize the dual-band SAR FPTD.
3) We further conduct airborne circular SAR (CSAR) [9] foliage-penetrating experiments to verify the wide applicability of DBIFFPTD in the curvilinear flight.

II. Background

A. Nonsubsampled Shearlet Transform

During the standard shearlet transform, filter banks are obtained by the transformation of the window function in the pseudo-Doppler coordinate, which results in a downsampling operation. Therefore, the standard shearlet transform does not have the shift-invariant which leads to a pseudo-Gibbs phenomenon [10] when reconstructing images. Nonsubsampled shearlet transform (NSST) [11] is an improvement over the shearlet transform where shearlet filter (SF) banks are mapped from the pseudo-Doppler coordinate to the Cartesian coordinate. Then, other steps can be done based on the 2-D convolution and the inverse Fourier transform (IFFT) to avoid the downsampling operation, thus NSST has the shift-invariant. NSST combines nonsubsampled pyramid filter (NSPF) banks with SF banks to provide several independent sparse representations to describe the image information more effectively. The flowchart of NSST is shown in Fig. 1 where the decomposition level of NSPF banks is set as 3.

Step 1: (Multilevel NSPF Decomposition): After the first-level NSPF decomposition, the source image is decomposed into a low-passband ${f_L}^1$ and a high-pass subband ${f_H}^1$. Then, the second-level NSPF decomposition is performed on the low-pass subband ${f_L}^1$ to obtain the low-pass subband ${f_L}^2$ and the high-pass subband ${f_H}^2$. Next, the third-level NSPF decomposition is performed on the low-pass subband ${f_L}^2$ to obtain the low-pass subband ${f_L}^3$ and the high-pass subband ${f_H}^3$. Finally, we can obtain one low-pass subband ${f_L}^3$ with three high-pass subbands $\{{f_H}^i\}_{i=1}^3$. The NSPF in each decomposition level is a two-channel 2-D filter shown in Fig. 2, where $F_L(z)$, $F_H(z)$, $G_L(z)$, and $G_H(z)$ represent the low-pass decomposition filter, the high-pass decomposition filter, the low-pass reconstruction filter, and the high-pass reconstruction filter, respectively. Besides, $F_L(z)$, $F_H(z)$, $G_L(z)$, and $G_H(z)$ satisfy the following Bezout constraints [12] as

$$F_L(z)G_L(z) + F_H(z)G_H(z) = 1. \tag{1}$$

Step 2: (Localization in Directions): The shearlet transform utilizes the scale transformation matrix to construct the anisotropic wavelet basis. For decomposing images, the shearlet transform must be discrete by selecting the proper scale parameter a, the shear parameter s, and the shift parameter t. The discrete shearlet function (DSF) [10] can be shown as follows:

$$\psi_{a,s,t} = \left|\det \mathbf{U}^{a/2}\right| \psi\left(\mathbf{S}^s \mathbf{U}^a x - t\right) \tag{2}$$

where $\mathbf{U} = [2, 0; 0, \sqrt{2}]$ is the anisotropic dilation matrix and $\mathbf{S} = [1, 1; 0, 1]$ is the shearlet matrix. Based on DSF, the localization in various directions of three high-pass subbands can be obtained and the specific process is as follows, that is, first DSF is generated by constructing window functions $\psi(\cdot)$-based on the "Meyer" wavelet, then DSF is mapped from the pseudo-Doppler coordinate to the Cartesian coordinate, next three high-pass subbands are convoluted with the DSF to obtain directional subbands, respectively, and finally directional subbands with shift-invariant are obtained by IFFT.

B. Nonnegative Matrix Factorization

Nonnegative matrix factorization (NMF) is an unsupervised matrix analytic method that decomposes a matrix with

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nonnegative elements. This method can iteratively extract main features from different images [13]. When the pixel value of the imageset is nonnegative, the abstract feature of the imageset can be obtained by setting the feature dimension.

Based on NMF, a large-scale matrix $\mathbf{V}_{M\times N}$ with M members and N-dimensional feature can be decomposed into two small-scale matrix $\mathbf{W}_{M\times R}$ and $\mathbf{H}_{R\times N}$ such that

$$\mathbf{V}_{M\times N} = \mathbf{W}_{M\times R}\mathbf{H}_{R\times N} + \varepsilon \tag{3}$$

where $\mathbf{W}$ is defined as the basis matrix with $R < (MN/(M+N))$, that is, the dimension of original features in $\mathbf{V}_{M\times N}$ is decreased to R, $\mathbf{H}$ is the coefficient matrix that approximately reconstructs the original N-dimension feature from the decreased R-dimension feature, and ε is the existing error. To minimize the error between $\mathbf{V}$ and $\mathbf{WH}$. The loss function should be defined to quantify the quality of error ε as the following expression:

$$\begin{aligned} L(\mathbf{W},\mathbf{H}) &= \|\mathbf{W}_{M\times R}\mathbf{H}_{R\times N}\|^2 \\ &= \sum_{m=1}^{M}\sum_{n=1}^{N}\left(\mathbf{V}_{m,n} - (\mathbf{WH})_{m,n}\right)^2 \end{aligned} \tag{4}$$

where the representative low-dimensional feature $\mathbf{W}_{M\times R}$ would make the loss function approach to zeros.

For optimizing the loss function in (4), Lee and Seung [14] proposed a multiplicative update rule to get the optimal NMF instead of the additive update rule. The content of the multiplicative update rule is shown in the following steps.

Step 1: Setting the partial derivative of (4) for $\mathbf{W}$ and $\mathbf{H}$ to zeros, respectively, as

$$\begin{cases} \dfrac{\partial L(\mathbf{W},\mathbf{H})}{\partial \mathbf{W}} = -\mathbf{V}\mathbf{H}^{\mathrm{T}} + \mathbf{W}\mathbf{H}\mathbf{H}^{\mathrm{T}} = 0 \\ \dfrac{\partial L(\mathbf{W},\mathbf{H})}{\partial \mathbf{H}} = -\mathbf{W}^{\mathrm{T}}\mathbf{V} + \mathbf{W}^{\mathrm{T}}\mathbf{W}\mathbf{H} = 0. \end{cases} \tag{5}$$

Step 2: Updating $\mathbf{W}$ and $\mathbf{H}$ iteratively as

$$\begin{cases} \mathbf{W}_{i+1} = \mathbf{W}_i \dfrac{\mathbf{V}{\mathbf{H}_i}^{\mathrm{T}}}{\mathbf{W}_i\mathbf{H}_i\mathbf{H}_i^{\mathrm{T}}} \\ \mathbf{H}_{i+1} = \mathbf{H}_i \dfrac{\mathbf{W}_{i+1}^{\mathrm{T}}\mathbf{V}}{\mathbf{W}_{i+1}\mathbf{H}_i\mathbf{H}_i^{\mathrm{T}}}. \end{cases} \tag{6}$$

Comparing with the following traditional gradient descent solution in the quadratic optimization:

$$\begin{cases} \mathbf{W}_{i+1} = \mathbf{W}_i - \eta\left(-\mathbf{V}{\mathbf{H}_i}^{\mathrm{T}} + \mathbf{W}_i\mathbf{H}_i\mathbf{H}_i^{\mathrm{T}}\right) \\ \mathbf{H}_{i+1} = \mathbf{H}_i - \eta\left(-\mathbf{W}_{i+1}^{\mathrm{T}}\mathbf{V} + \mathbf{W}_{i+1}\mathbf{H}_i\mathbf{H}_i^{\mathrm{T}}\right) \end{cases} \tag{7}$$

where η is the step size, (6) can guarantee the sparse representation of $\mathbf{V}$ is nonnegative, and thus NMF is widely used in image processing.

C. Modified Laplacian Energy

The sum of modified Laplacian energy (SML) [15] is a method that can appropriately reflect edges, textures, and orientation information of images. The classical Laplacian energy (LPE) can be defined as

$$\begin{aligned} \mathrm{LPE}(x,y) &= |2\mathbf{C}(x,y) - \mathbf{C}(x-k,y) - \mathbf{C}(x+k,y)| \\ &\quad + |2\mathbf{C}(x,y) - \mathbf{C}(x,y-k) - \mathbf{C}(x,y+k)| \end{aligned} \tag{8}$$

where k denotes the variable step and $\mathbf{C}(x,y)$ is the pixel value of the image at position x and y. Then, SML could be further defined as the sum of the local LPE in (8), that is,

$$\mathrm{SML}(x,y) = \sum_{a=-M}^{M}\sum_{b=-N}^{N}\left[LP(x+a,y+b)\right]^2 \tag{9}$$

where $(2M+1)\times(2N+1)$ is the size of the sliding window. Besides, the pixels in diagonal directions could also be considered to rewrite (8) as

$$\begin{aligned} &\mathrm{LPE}(x,y) \\ &= |2\mathbf{C}(x,y) - \mathbf{C}(x-k,y) - \mathbf{C}(x+k,y)| \\ &\quad + |2\mathbf{C}(x,y) - \mathbf{C}(x,y-k) - \mathbf{C}(x,y+k)| \\ &\quad + \frac{|2\mathbf{C}(x,y) - \mathbf{C}(x+k,y-k) - \mathbf{C}(x-k,y+k)|}{\sqrt{2}} \\ &\quad + \frac{|2\mathbf{C}(x,y) - \mathbf{C}(x+k,y+k) - \mathbf{C}(x-k,y-k)|}{\sqrt{2}}. \end{aligned} \tag{10}$$

III. Framework of the Proposed FPTD Method

The single-pass dual-band SAR change information is an effective feature to detect FP targets. The core of extracting dual-band SAR change features is to transfer the texture of high-band SAR images to low-band SAR images to get fake high-band SAR images. The transferred fake high-band SAR image would obtain similar texture information to the original high-band SAR image. However, the transferred result by the existing CGAN method suffers from image discontinuities and a heavy time complexity.

In the proposed novel dual-band SAR image-based FPTD method, we utilize SAR image fusion to obtain the fake high-band SAR image. Dual-band image fusion in this article decomposes dual-band SAR images by NSST, fuses low-pass subbands by NMF, and fuses high-pass subbands by SML. The difference operator is utilized to measure the high-band SAR image and the fake high-band SAR image. Besides, the threshold method is taken to detect FP targets. The processing flow is outlined in Fig. 3.

A. Dual-Band SAR Image Fusion

NSST is a new multiscale geometric analysis tool that has the shift-invariant. Besides, NSST can obtain a compactly supported frame that has no limit on the size of support areas and the number of directions during the shearlet process. Thus, NSST can make a more detailed description of images. Dual-band SAR images would be fused based on NSST as follows.

Low-Pass Subbands Fusion: Applying NSST to decompose source images $\mathbf{A}$ and $\mathbf{B}$ which have been accurately registered, corresponding low-pass subbands would be obtained as $\boldsymbol{f}_L^{\mathbf{A}^l}$, $\boldsymbol{f}_L^{\mathbf{B}^l}$, where l means the number of the decomposition level, L is the mark of low-pass subbands, and $\boldsymbol{f}$ means the column vector form of the image matrix. For the DBIF, the matrix $\mathbf{V}$ in NMF can be set as $\mathbf{V} = [\boldsymbol{f}_L^{\mathbf{A}^l}; \boldsymbol{f}_L^{\mathbf{B}^l}]$ and the number of columns of $\mathbf{W}$ is set as 1. After iterations, $\mathbf{W}$ contains the main features of two source images $\mathbf{A}$ and $\mathbf{B}$.

High-Pass Subbands Fusion: Since SML can better reflect the consequence of the high-pass subband, that is, the abundance of the edge and texture information can be expressed

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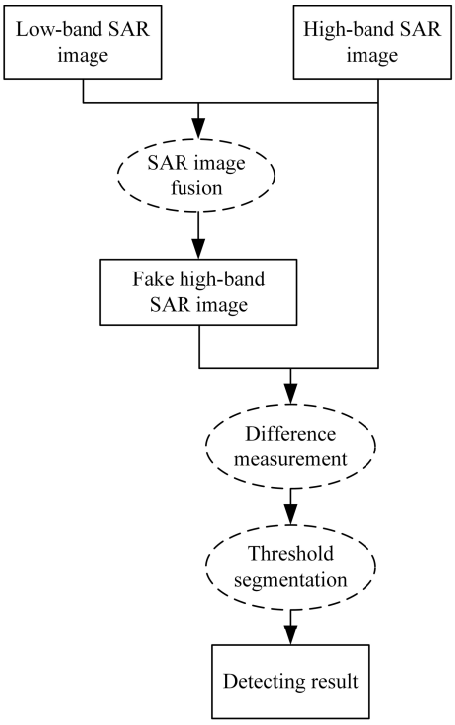


Fig. 3. Flow of the proposed FPTD method.

by the LPE, the maximal SML is adopted to fuse the two high-pass subbands of the source image **A** and **B** as

$$f_{\text{Fuse}}^{l,d}(x,y)=\begin{cases} f_{H}^{\mathbf{A}\,l,d}(x,y), & \text{SML}_{\mathbf{A}}^{l,d}(x,y)\geq \text{SML}_{\mathbf{B}}^{l,d}(x,y) \\ f_{H}^{\mathbf{B}\,l,d}(x,y), & \text{SML}_{\mathbf{A}}^{l,d}(x,y)\leq \text{SML}_{\mathbf{B}}^{l,d}(x,y) \end{cases} \tag{11}$$

where H is the mark of high-pass subbands, d is the direction of the shearlet filter, and $f_{\text{Fuse}}^{l,d}(x,y)$ means the merged subband at the decomposition level l, and direction d. Thus, $f_{\text{Fuse}}^{l,d}(x,y)$ contains both details of source images **A** and **B**.

B. Difference Measurement

To measure the difference between the high-frequency SAR image and the fake high-frequency SAR image, the intensity difference between the image pair as a statistic criterion [16] can be written as

$$\left|\sum_{a=-M}^{M}\sum_{b=-N}^{N}\mathbf{B}(x+a,y+b)^2-\sum_{a=-M}^{M}\sum_{b=-N}^{N}\hat{\mathbf{B}}(x+a,y+b)^2\right| \tag{12}$$

where $\hat{\mathbf{B}}$ is the approximated **B**. Besides, (12) constructs the statistics criterion based on the intensity difference of the corresponding $(2M+1)\times(2N+1)$ neighborhood pixels in the image pair.

C. Threshold Segmentation

The OTSU threshold method [17] is commonly used to binarize the image. As an adaptive threshold confirmation method, the image can be segmented into the target object and background region according to the maximum interclass variance of the image gray level. After reasonably calculating the maximum variance between classes, the optimal threshold can be obtained.

Set the gray level of the image as $0\sim 255$, grade r pixel contains s_r pixel blocks, and $S=\sum_{r=0}^{255}s_r$ is the total pixel, thus the probability of grade r level is set as $p_r=s_r/S$. When $r=\mu$ is used as the gray threshold value, image pixels can be divided according to the background region is $\mathbf{C}_B=0\sim\mu$ and the target object is $\mathbf{C}_O=\mu+1\sim 255$. Thus, the probability of $\mathbf{C}_B$ and $\mathbf{C}_O$ being presented is as follows:

$$\begin{cases} Pr_{\mathbf{C}_B}=\sum_{r=0}^{\mu}p_r=\omega(\mu) \\ Pr_{\mathbf{C}_O}=\sum_{r=\mu+1}^{255}p_r=1-\omega(\mu) \end{cases} \tag{13}$$

where $\omega(\mu)$ is the gray level calculation coefficient.

μ from $0\sim 255$ which leads to the maximum value of $\sigma^2(\mu)$ is the optimal threshold as follows:

$$\mu^*=\arg\max_{0<\mu<255}\left[\sigma^2(\mu)\right] \tag{14}$$

where $\sigma^2(\mu)$ is the sum of interclass variances of the two different regions which are divided by μ.

According to the calculation result of the best threshold in (14), we can judge whether the false alarm is alleviated. Besides, the multithreshold OTSU [18] as a member of the OTSU family could extract the target area more accurately than the binary OTSU in (14) at times.

IV. Theory Analysis and Simulation

To theoretically verify the feasibility of the proposed fake high-band SAR image generation method by the DBIF, we first discuss the scattering of radar waves reflected from the surface with various degrees of roughness (the distance of the adjacent peak and valley in height). Then, we discuss the scattering characteristic and the intensity between various landscape objects (water, soil, crops, forests, and FP targets) in dual-band SAR images from the aspect of the roughness, target shape, and foliage penetration. Finally, we simulate dual-band SAR images which contain various landscape objects and generate the simulated fake high-band SAR image by the DBIF.

A. Reflection Forms Analysis

For the electromagnetic wave, the reflection consists of three forms: specular reflection, diffuse reflection, and directional reflection. Fig. 4(a) demonstrates that the specular reflection on the absolutely smooth surface is strictly directional, that is, the incidence angle θ is equal to the reflection angle. Then, Fig. 4(b) presents that the diffuse reflection on the absolutely rough surface is isotropic, that is, the electromagnetic wave is scattered uniformly to the space, thus the reflection direction is irrelevant to the incidence direction. Besides, Fig. 4(c) illustrates that the directional reflection on the moderate rough surface is nonuniform and has strong scatters in certain directions. For Fig. 4(b) and (c), the degree of surface roughness can be indirectly reflected by the height difference h between the local peak and valley on the surface.

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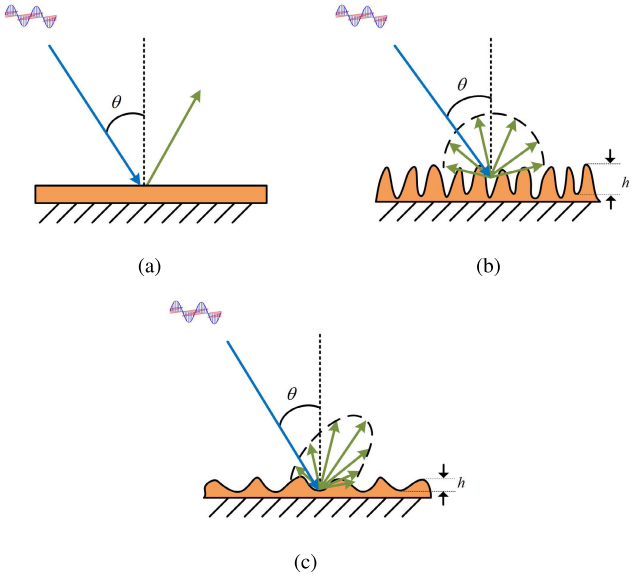


Fig. 4. Reflection forms of the electromagnetic wave from different surfaces. (a) Specular reflection. (b) Diffuse reflection. (c) Directional reflection.

Based on the improved Rayleigh criterion [19], we can learn that the absolute smooth and absolute rough surface are commonly nonexistent on the ground surface but the specular reflection and the diffuse reflection can be approximately generated from the directional reflection by the radar wave with various wavelengths λ and incidence angles θ, thus the roughness of the reflecting surfaces is relative to the electromagnetic wave. Fig. 5(a) and (b) demonstrates the directional reflection generated by the moderate rough surface at

$$\begin{cases} 4.4\ \mathrm{h}\cos\theta < \lambda < 25\ \mathrm{h}\cos\theta \\ \cos^{-1}\left(\dfrac{\lambda}{4.4\ \mathrm{h}}\right) < \theta < \cos^{-1}\left(\dfrac{\lambda}{25\ \mathrm{h}}\right) \end{cases} \tag{15}$$

respectively. After increasing the wavelength and incidence angle, the original directional reflection in Fig. 5(a) and (b) would, respectively, change to the quasispecular reflection in Fig. 5(c) and (d). The conditional wavelength and incidence angle should satisfy

$$\begin{cases} \lambda > 25\ \mathrm{h}\cos\theta \\ \theta > \cos^{-1}\left(\dfrac{\lambda}{25\ \mathrm{h}}\right) \end{cases} \tag{16}$$

respectively. On the contrary, the original directional reflection in Fig. 5(a) and (b) would, respectively, change to the quasidiffuse reflection showed in Fig. 5(e) and (f) by decreasing the wavelength and incidence angle to

$$\begin{cases} \lambda < 4.4\ \mathrm{h}\cos\theta \\ \theta < \cos^{-1}\left(\dfrac{\lambda}{4.4\ \mathrm{h}}\right) \end{cases} \tag{17}$$

respectively. Based on (15)–(17), the scattering characteristic of the ground surface can be roughly deduced by the reflection form.

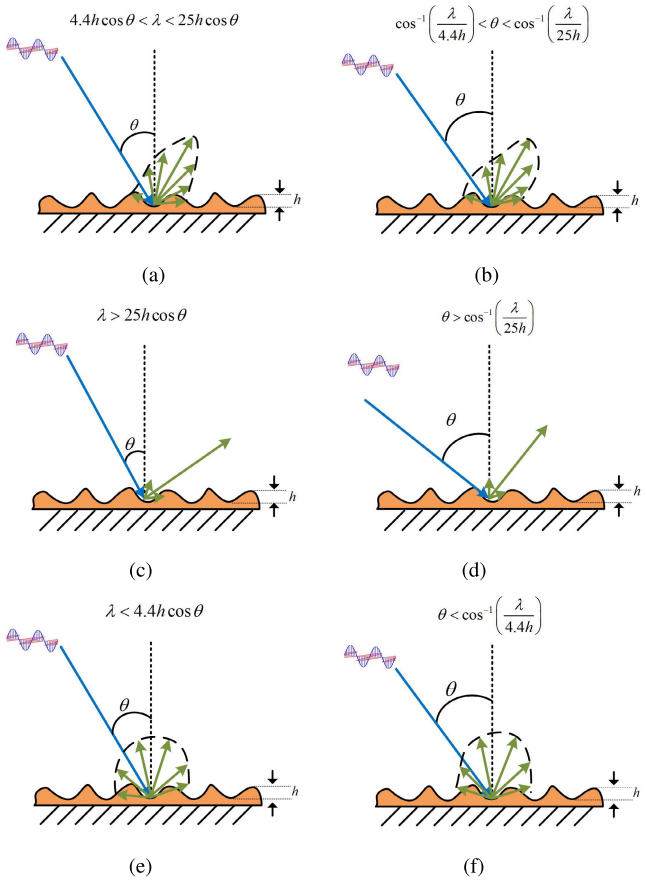


Fig. 5. Reflection forms of the electromagnetic wave with different wavelengths and incidence angles from the moderate rough surface. (a) and (b) Directional reflection. (c) and (d) Quasispecular reflection. (e) and (f) Quasidiffuse reflection.

B. Landscape Object Analysis

In accordance with practice, the radar wave is equal to or longer than an L-band radar wave in wavelength ($\lambda \geq$ 22 cm) is defined as the low-band radar wave, while the radar wave being equal or shorter than an S-band radar wave in wavelength ($\lambda \leq$ 10 cm) is defined as the high-band radar wave [20]. Ku-band ($\lambda \approx$ 2 cm) and P-band ($\lambda \approx$ 65 cm) radar waves are utilized in the theory analysis and incidence angles are both assumed to be $\theta \approx (\pi/4)$, thus we would deduce scattering characteristics of the landscape object and analysis the scattering intensity of SAR results.

1) Water Area: We assume the degree of the water roughness is about 0.1 cm at the calm area, that is, $h < 0.1$ cm. For the Ku-band and P-band radar wave, λ, θ, and h satisfy (16), thus the reflection form at the calm water is the quasispecular reflection. Fig. 6(a) and (b) illustrates that the radar wave hardly reflected the radar direction, thus the water area has a weak backscattering and presents a low intensity in the Ku-band and P-band SAR image.

2) Soil Area: We assume that the degree of the soil roughness is about 2 cm in the bare area. For the Ku-band and P-band radar wave, λ, θ, and h satisfy (17) and (16), respectively, thus the Ku-band and P-band radar wave would, respectively, generate the quasidiffuse reflection and quasispecular reflection at the bare soil. Fig. 6(c) illustrates that the

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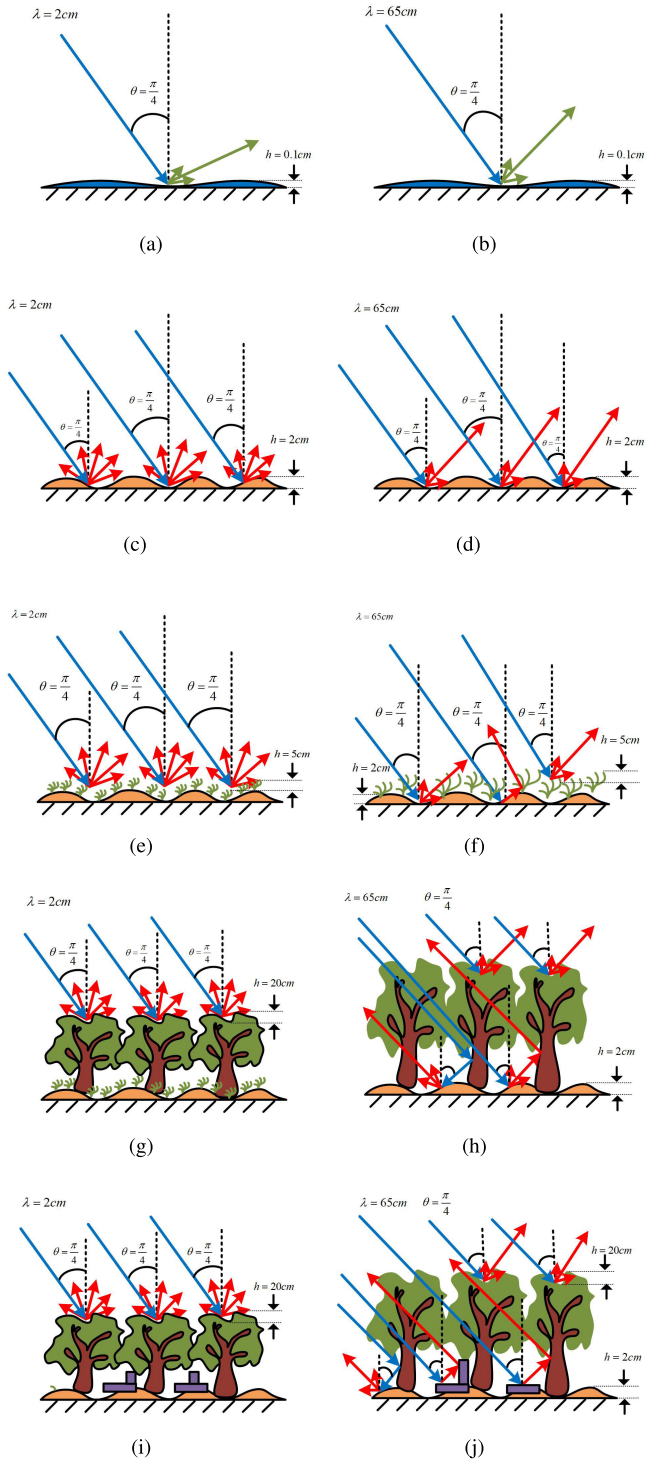


Fig. 6. Reflection forms of the electromagnetic wave with different wavelengths and incidence angles from the moderate rough surface. (a) and (b) Directional reflection. (c) and (d) Quasispecular reflection. (e)–(j) Quasidiffuse reflection.

quasidiffuse reflection would lead to a strong backscattering in the radar direction, thus the soil area would have a higher intensity than the water area in the Ku-band SAR image. On the contrary, the quasispecular reflection leads to a weak backscattering at the radar direction as illustrated in Fig. 6(d), thus the soil area in the P-band SAR image is also low intensity. Besides, the soil area would be covered with ruderal weeds, thus the weed-covered soil area has a slightly lower intensity than the bare soil area in the SAR image for the absorption of radar waves.

3) Crop Area: According to the theory of FP, we learn that the Ku-band radar wave would be absorbed by the multiple reflections in the crop canopy, while the P-band radar wave could directly penetrate the crop canopy to the bare soil with a slight attenuation. We assume the degree of the crop roughness is about 5 cm at the crop canopy and is about 2 cm at the soil under the crop canopy. For the Ku-band radar wave in Fig. 6(e), λ, θ, and h satisfy (17), thus the reflection form at the surface of the crop canopy is the quasidiffuse reflection and the intensity of crop areas would be slightly lower than it of soil areas in the Ku-band SAR image for the absorption of radar waves by the crop canopy. For the P-band radar wave in Fig. 6(f), λ, θ, and h satisfy (16) at the crop canopy, thus the reflection form at the crop canopy is the quasispecular reflection. Besides, P-band radar waves would mainly penetrate the crop canopy and have a quasispecular reflection at the soil except few radar waves would have a directional reflection at the crop canopy. For radar waves reflected by the soil, one part would have a secondary reflection to the radar direction by the crop canopy, the other part would directly have a secondary penetrating through the crop canopy, thus the crop area has a slightly higher intensity than it of soil areas in the P-band SAR image.

4) Forest Area: We assume the roughness degree at the forest canopy is about 20 cm which is much larger than it is at the crop canopy. For the Ku-band radar wave in Fig. 6(g), λ, θ, and h satisfy (17), thus the reflection form at the forest canopy is the quasidiffuse reflection and the forest area in the Ku-band SAR image would have a high intensity. Then, thick leaves would make the forest canopy more enclosed than the crop canopy so that the Ku-band radar wave could hardly reach the interior of the forest canopy, thus the intensity of forest areas would be higher than that of crop areas and weeds-covered soil areas in the Ku-band SAR image for the few absorption of radar waves by the forest canopy. Besides, the tree canopy is larger than the crop canopy, thus the outline of the forest canopy would be distinct in the Ku-band SAR image. For the P-band radar wave in Fig. 6(h), λ, θ, and h satisfy (16) at the forest canopy, thus the reflection form at the forest canopy is the quasispecular reflection. Like the crop area, the radar wave penetrating the canopy has a quasispecular reflection on the soil. However, most of the radar wave reflected by the soil would have a secondary reflection to the radar direction at the trunk because the trunk of forests is much thicker than the stalk of crops, thus the forest area has a higher intensity than the crop area in the P-band SAR image without a distinct outline of the forest canopy.

5) Foliage-Penetrating Target: Military FP targets are commonly cars and weapons with a metal surface and regular shape, thus we can simplify targets by the metal dihedral corner and the metal plane. For the Ku-band radar wave in Fig. 6(i), the forest canopy cannot be penetrated. Thus, FP targets cannot be highlighted among the imaging results of

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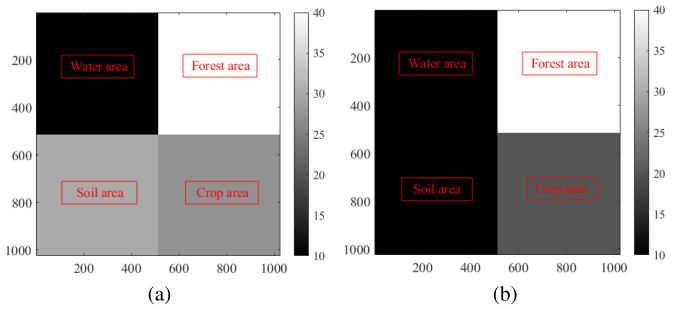


Fig. 7. Assumed scattering coefficient images for the dual-band SAR. (a) Ku-band SAR. (b) P-band SAR.

forest canopies and the forest area has a high intensity in the Ku-band SAR image. For the P-band radar wave in Fig. 6(j), the penetrated wave would be reflected by the dihedral-corner target, metal plane, and thick trunk. Compared with the soil under the forest, the radar wave almost has no absorption on the metal surface, thus the received radar wave in Fig. 6(j) is stronger than it in Fig. 6(h) and the FP target would have a higher intensity than it of the forest area in the P-band SAR image.

C. Simulated Result Analysis

To theoretically verify the feasibility of the proposed fake high-band SAR image generation method, dual-band SAR images (Ku-band and P-band) with the size of 1024×1024 are simulated based on the scattering characteristic analysis and the backscattering intensity analysis of landscape objects.

We first assume the desired scattering coefficient s_{ku} and s_p indicated in Fig. 7(a) and (b) where the 1024×1024 images are divided into four parts, that is, water area ($s_{ku} = 2$, $s_p = 2$), weed-covered soil area ($s_{ku} = 30, s_p = 2$), crop area ($s_{ku} = 27, s_p = 20$), and forest area ($s_{ku} = 40$, $s_p = 40$). Then, we assume the multiplicative speckle noise [22] of dual-band SAR images obeying the zeros-mean Rayleigh distribution, and the two standard deviations are defined as σ_{ku} and σ_p. Besides, the intensity of the speckle noise would be enhanced as the increase of the roughness degree [23], that is, the standard deviation would be large for the quasidiffuse surface and would be small for the quasis-pecular surface. Water area ($\sigma_{ku} = 2, \sigma_p = 2$), weeds-covered soil area ($\sigma_{ku} = 6, \sigma_p = 3.6$), crop area ($\sigma_{ku} = 5.4, \sigma_p = 4$), and forest area ($\sigma_{ku} = 8, \sigma_p = 8$). Adding the multiplicative simulated Rayleigh noise [24], we can obtain two simulated dual-band SAR images in Fig. 8(a) and (b) where small-size FP targets with smooth metal surfaces are almost noiseless. Finally, the dual-band SAR fusion image indicated in Fig. 8(c) can be obtained by fusing Fig. 8(a) and (b). Besides, Fig. 9 shows the similarity comparison of the dual-band SAR fusion image that is, the structural similarity (SSIM) and probability density functions (pdfs) of dual-band SAR images where SSIM represents the similarity of the co-registered pixel, the global SSIM is the mean SSIM of all co-registered pixels, and pdf means the global distribution of pixels. Smaller global SSIM would show lower pixel values in Fig. 9(a) and (b).

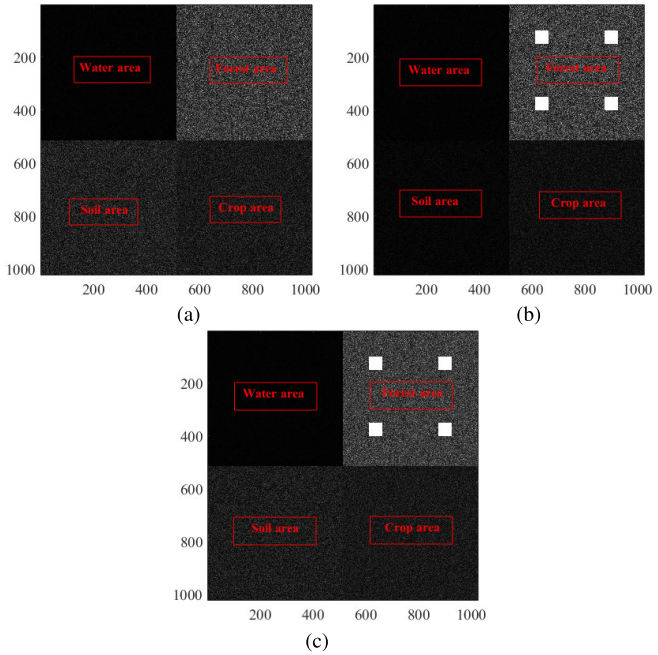


Fig. 8. Simulated dual-band SAR images with the Rayleigh noise and the dual-band fusion result. (a) Ku-band SAR. (b) P-band SAR. (c) Dual-band fusion SAR.

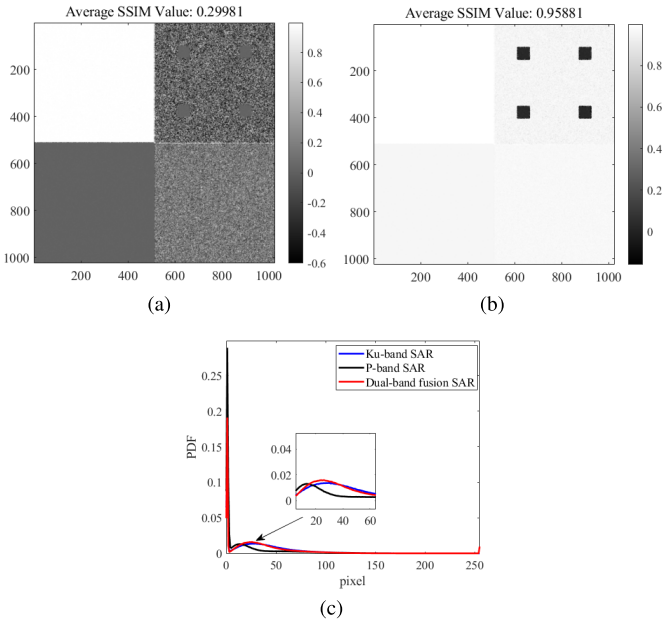


Fig. 9. Similarity comparison of the dual-band SAR fusion image and dual-band SAR images. (a) SSIM image of Fig. 8(a) and (b). (b) SSIM image of Fig. 8(a) and (c). (c) Pixel distribution.

We can learn from Fig. 9 that the texture similarity of Fig. 8(a) and (c) is higher than it of Fig. 8(a) and (b), then the pixel distribution of Fig. 8(a) is similar with it of Fig. 8(c) rather than Fig. 8(b). Since the dual-band SAR fusion image retains the FP target shown in the P-band SAR image, we consider that the dual-band SAR fusion method can be utilized to generate the fake high-band SAR image by transferring the texture information from high-band SAR images to low-band SAR images.

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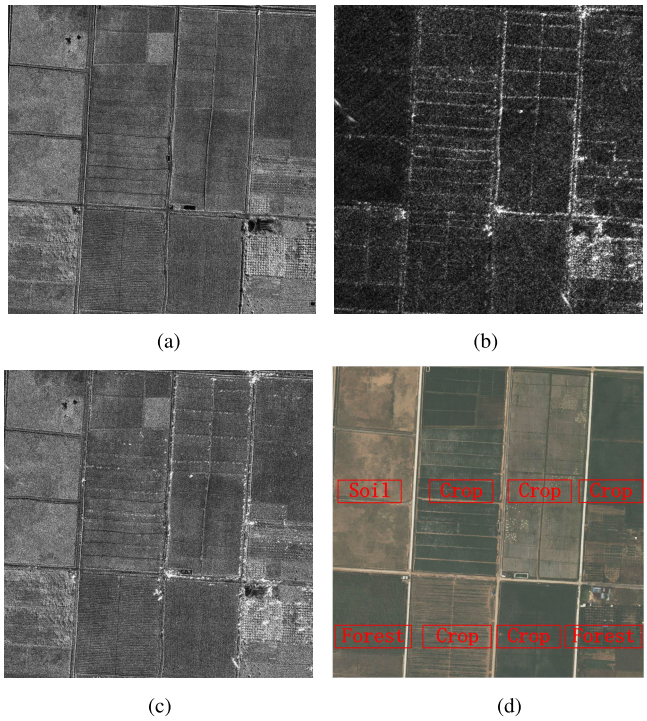


Fig. 10. Dual-band LSAR images and fake Ka-band LSAR image of farmland in Wanning city, China. (a) Ka-band LSAR. (b) P-band LSAR. (c) Fake Ka-band LSAR by DBIF. (d) Optics.

V. Experiment and Analysis

In this section, co-registered dual-band linear SAR (LSAR) images are selected to present the superiority of the dual-band LSAR image fusion [8]-based texture transferring method and the feasibility of the DBIFFPTD method. The traditional double-parameter constant false alarm rate (DCFAR) detector-based FPTD (DCFARFPTD) [4] on the low-band LSAR image and the CGAN-based FPTD (CGANFPTD) [5] on the low-band LSAR image and high-band LSAR image are chosen for the performance comparison with the proposed DBIFFPTD in the time complexity and detection result. Besides, we also conduct the dual-band CSAR experiment to show the wide applicability of DBIFFPTD.

A. Texture Transferring Analysis

The effectiveness of the proposed DBIF-based texture information transferring method is verified by HV-mode linear AIR-MD-SAR-Map data [21] obtained from the Chinese aeronautic remote sensing system (CARSS) and our independent measured HH-mode LSAR data. CGAN with the 512×512 input as a traditional LSAR texture information transferring method is selected as a comparison. Besides, SSIM [25] and pdf [26] are selected to measure the quality of fake high-band SAR images by CGAN and DBIF.

1) AIR-MD-SAR-MAP: Co-registered dual-band (Ka-band and P-band) LSAR image patches from AIR-MD-SAR-MAP with size 2560×2560 are shown in Fig. 10. We can see that the LSAR imaging scene is farmland containing the weeds-covered soil area, the crop area, and the forest area. Besides, dual-band LSAR images correspond to the landscape

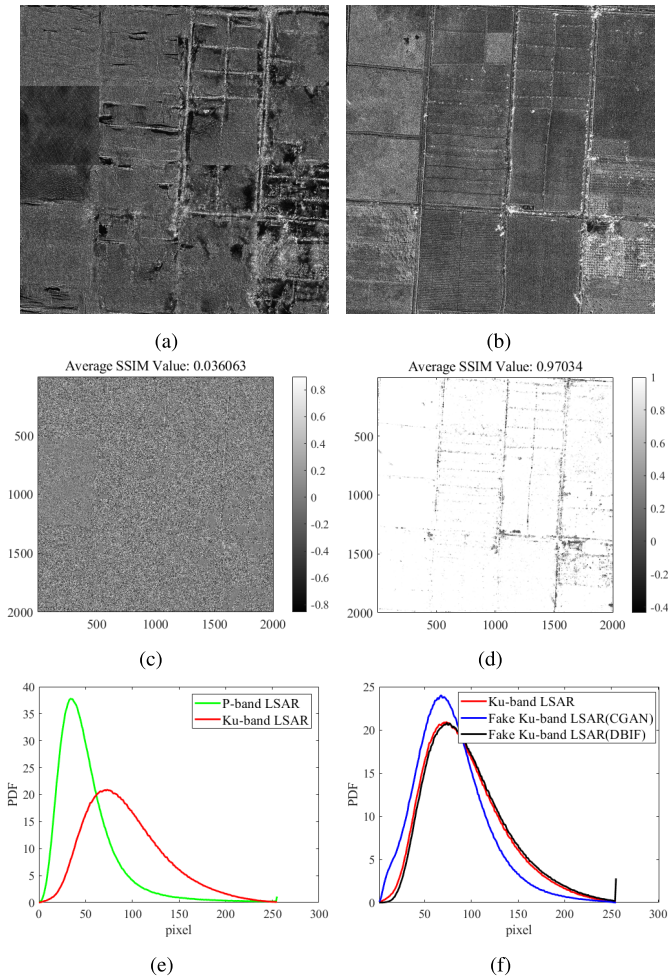


Fig. 11. Comparison of fake LSAR images by different texture transferring methods. (a) Fake Ka-band LSAR by CGAN. (b) Fake Ka-band LSAR by DBIF. (c) SSIM image of Fig. 10(a) and Fig. 11(a). (d) SSIM image of Fig. 10(a) and Fig. 11(b). (e) Pixel distribution. (f) Pixel distribution.

object analysis and the simulated dual-band SAR image, that is, the weeds-covered soil area has a higher intensity than the crop area and the forest area has a higher intensity than the crop area in the Ka-band LSAR image, while the weeds-covered soil area has a lower intensity than it of the crop area and the crop area has a lower intensity than it of the forest area in the P-band SAR image.

According to the optical image of SAR image patches, we can learn that the Ka-band LSAR image would present the texture information, for example, the outline of forests and crops, while the P-band SAR image would present edge information, for example, the ridge of fields. The fake Ka-band LSAR image by DBIF keeps the texture information which is the main body of the Ka-band LSAR image and enhances unapparent edge information by the P-band LSAR image.

Fig. 11 compares the quality of fake Ka-band LSAR images by the traditional CGAN method and the proposed DBIF method. We can see that Fig. 11(a) and (c) shows a low average SSIM value 0.036, while Fig. 11(b) and (d) shows a high average SSIM value 0.97. Comparing Fig. 11(c) and (d), we can see that most co-registered pixels present high SSIM values ($\approx$1) in the Ka-band LSAR image and the DBIF-based

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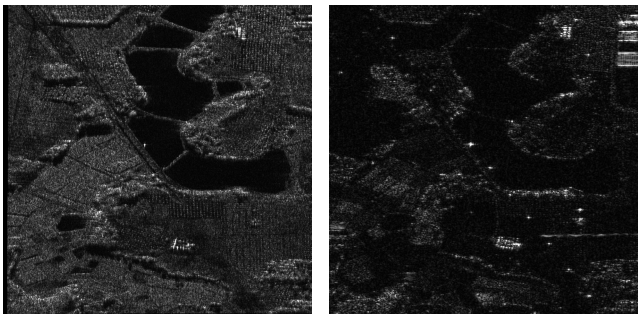


Fig. 12. Dual-band LSAR images of a wetland. (a) Ku-band LSAR. (b) P-band LSAR.

fake Ka-band LSAR image, then present low SSIM values (≈0) in the Ka-band LSAR image and the CGAN-based fake Ka-band LSAR image, thus the DBIF-based fake Ka-band LSAR image obtains abundant texture information. We can also see from Fig. 11(e) and (f) that the DBIF-based fake Ka-band LSAR image and the Ka-band LSAR image have more similar pixel distribution than the CGAN-based fake Ka-band LSAR image and Ka-band LSAR image. From the analysis of the texture information and the pixel distribution, we can conclude that DBIF has a better performance than CGAN in transferring the P-band LSAR image to the Ka-band LSAR image, that is, DBIF can generate a high-quality fake high-band SAR image from the low-band SAR image.

2) Our Measured Data: Co-registered dual-band (Ku-band and P-band) LSAR image patches with size 1024×1024 were recorded by our developed LSAR system in this article. As LSAR images shown in Fig. 12, the imaging scene is a wetland with a water area and vegetation area. We can see from Fig. 12 that the water area shows a relatively small pixel value and the vegetation area being similar to forests shows a relatively large pixel value. Besides, the P-band LSAR image contains several strong-scattering point targets which are slightly shown in the Ku-band LSAR image.

Fig. 13(a) and (b) illustrates fake Ku-band LSAR images transferred by CGAN and DBIF, respectively. We can see that the weight coefficient trained by CGAN is appropriate to transfer the vegetation area which has texture changes in the dual-band SAR image, while this trained weight is unsuitable for the water area which has similar pixel values in dual-band LSAR images, thus CGAN generates mistakes and discontinuities in Fig. 13(a). Comparing with Fig. 13(a), Fig. 13(b) shows an excellent continuity. We also further compare the quality of Fig. 13(a) and (b) with Fig. 12(a) by their SSIM values. We can see from 13(c) and (d) that Figs. 12(a) and (a) show a low average SSIM value of 0.229, while Figs. 12(b) and 13(b) show a high average SSIM value 0.915. SSIM-value images of co-registered pixels illustrate that Fig. 13(b) transfers more precise texture information than Fig. 13(a) from Fig. 12(a). Besides, we compare the quality of Fig. 13(a) and (b) by the pixel distribution. It can be seen from Fig. 13(e) and (f) that compared with the CGAN-based fake Ku-band LSAR image, the DBIF-based fake Ku-band LSAR image has a more similar pixel-value distribution with

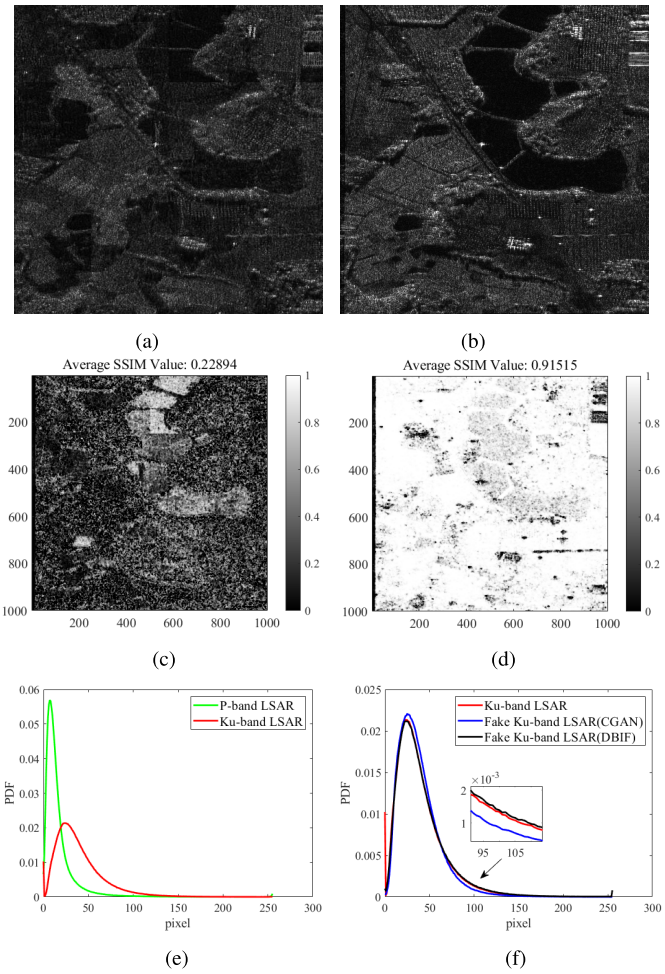


Fig. 13. Comparison of fake LSAR images by different texture transferring methods. (a) Fake Ku-band LSAR by CGAN. (b) Fake Ku-band LSAR by DBIF. (c) SSIM image of Fig. 12(a) and Fig. 13(a). (d) SSIM image of Fig. 12(b) and Fig. 13(b). (e) Pixel distribution. (f) Pixel distribution.

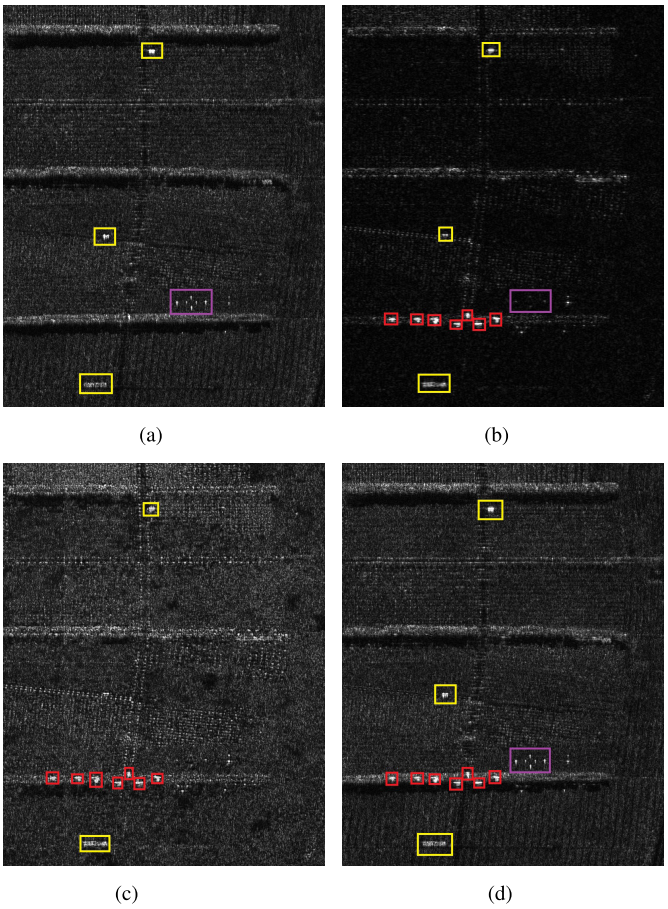
the original Ku-band LSAR image. Thus, we can conclude that DBIF has a better performance than CGAN in transferring the P-band LSAR image to the Ku-band LSAR image.

B. Detection Result Analysis

To evaluate the robustness of our proposed DBIFFTD, we measure the FP car and the FP street lamp by our developed dual-band (Ku-band and P-band) LSAR system and dual-band (Ku-band and L-band) CSAR system, respectively. DBIFFTD aims at detecting FP targets accurately with few false alarms under moderate time complexity.

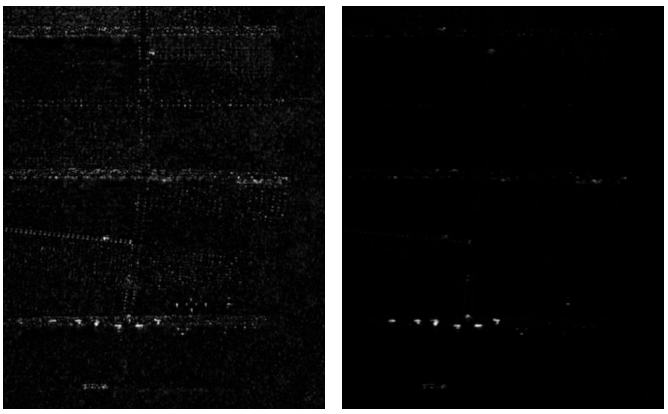
1) LSAR System: The observation scene in Fig. 14 is farmland with seven cars hidden under trees. Comparing Fig. 14(a) and (b), we can see that covert cars under trees can just be presented in the P-band LSAR image, that is, red-frame marks and exposed large strong-scattering targets can be presented in the dual-band LSAR images, that is, yellow-frame marks. Besides, exposed small strong-scattering targets are intensely presented in the Ku-band LSAR image and slightly presented in the P-band LSAR image, that is, the purple-frame mark. For DCFARFPTD which detects FP targets from the P-band LSAR image by the constant false alarm rate

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(a) (b) (c) (d)

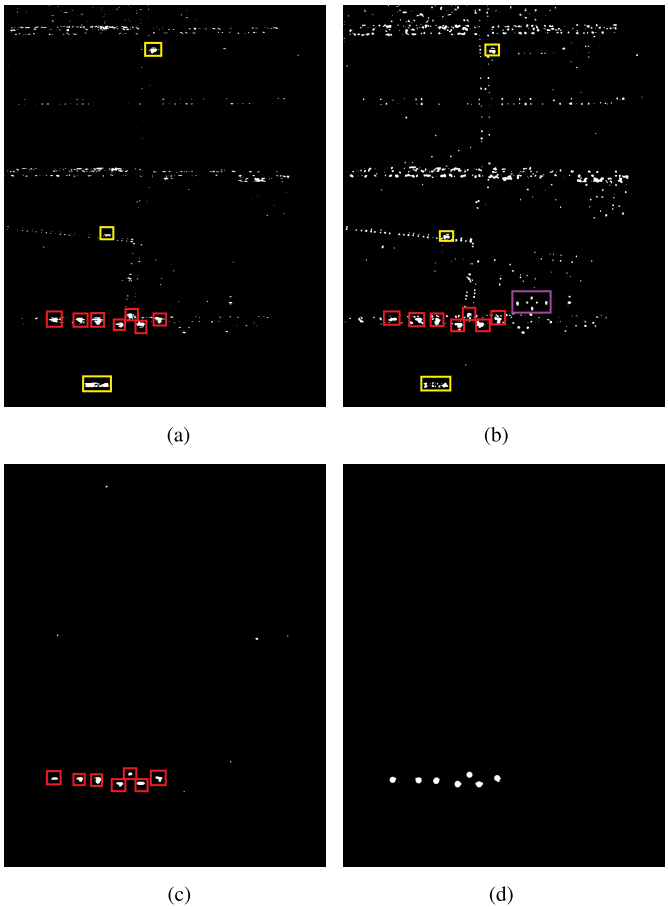
Fig. 14. Dual-band LSAR images and fake LSAR images of seven FP cars. (a) Ku-band LSAR. (b) P-band LSAR. (c) Fake Ku-band LSAR by CGAN. (d) Fake Ku-band LSAR by DBIF.



(a) (b)

Fig. 15. Difference image of the Ku-band and fake Ku-band LSAR images. (a) Difference image by CGANFPTD. (b) Difference image by DBIFFPTD.

(CFAR) method, those exposed strong-scattering targets with the similar size of FP targets in the yellow frame of Fig. 14(b) would generate false alarms in the detection result. For CGANFPTD which detects FP targets from Fig. 14(a) and (c), CGAN cannot transfer some exposed small strong-scattering targets from the Ku-band LSAR image to the P-band LSAR image, that is, targets in the purple frame, thus the difference image of Fig. 14(a) and (c) shown in Fig. 15(a) generates



(a) (b) (c) (d)

Fig. 16. Detection result of seven covert cars and ground truth. (a) Detection result by DCFARFPTD. (b) Detection result by CGANFPTD. (c) Detection result by DBIFFPTD. (d) Ground truth of targets.

undesirable intensity differences. Besides, the texture difference between Fig. 14(a) and (c) as disturbances are also added to the difference image. For DBIFFPTD which detects FP targets from Fig. 14(a) and (d), DBIF transfers exposed small strong-scattering targets and texture information accurately from the Ku-band LSAR image to the P-band LSAR image accurately, thus the difference image of Fig. 14(a) and (d) shown in Fig. 15(b) generates covert targets with few disturbances, that is, yellow-frame exposed large strong-scattering targets and purple-frame small strong-scattering targets are completely eliminated by the difference operator.

Fig. 16 illustrates the detection result of DCFARFPTD, CGANFPTD, and the proposed DBIFFPTD from Figs. 14(b) and 15(a) and (b), respectively. Considering the difference image [Fig. 15(b)] has a few false alarms, and the efficient OSTU method is taken to directly detect FP cars. We can learn from Fig. 16 that DCFARFPTD and CGANFPTD cannot overcome the disturbance of both purple-frame and yellow-frame false alarms from exposed strong-scattering targets with different sizes. Besides, the CGANFPTD still suffers from the texture difference caused by the training weight of CGAN. However, DBIFFPTD could detect seven FP cars accurately with a few small-size false alarms. Compared with CGANFPTD which transfers the texture information ahead by the time-consuming CGAN, DBIFFPTD replaces the CGAN

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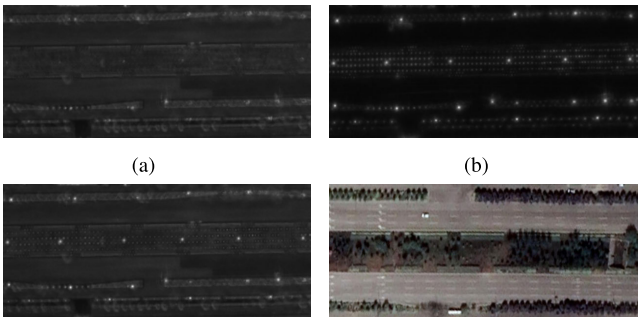


Fig. 17. Dual-band CSAR images and fake CSAR images of various street lamps in the first scene. (a) Ku-band CSAR. (b) L-band CSAR. (c) Fake Ku-band CSAR by DBIF. (d) Optics.

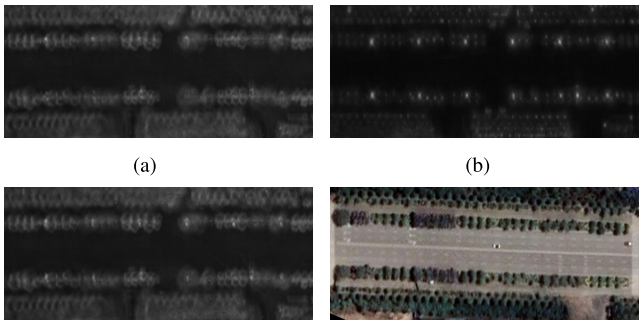


Fig. 19. Dual-band CSAR images and fake CSAR images of various street lamps in the second scene. (a) Ku-band CSAR. (b) L-band CSAR. (c) Fake Ku-band CSAR by DBIF. (d) Optics.

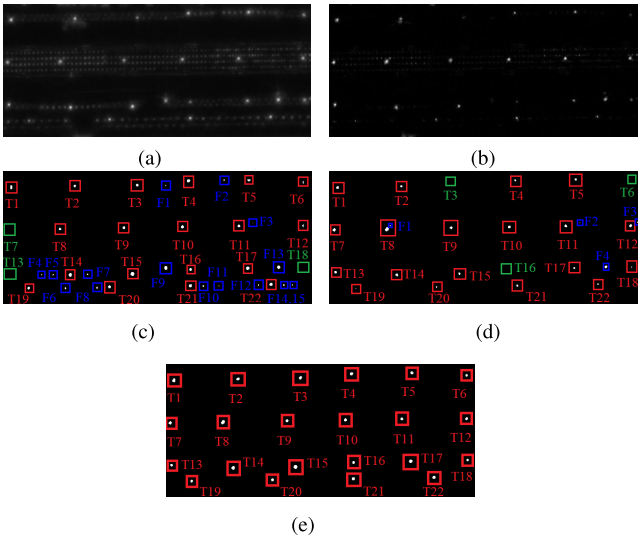


Fig. 18. Detection result of covert street lamps and ground truth in the first street scene. (a) L-band CSAR. (b) Difference image by DBIFFPTD. (c) Detection result by DCFARFPTD. (d) Detection result by DBIFFPTD. (e) Ground truth.

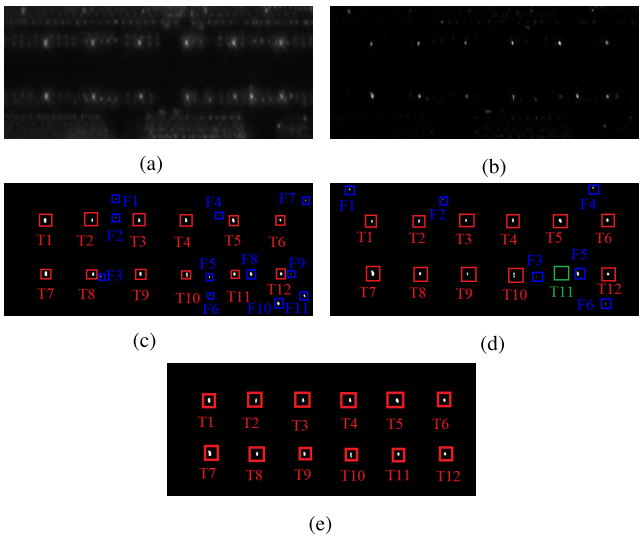


Fig. 20. Detection result of covert street lamps and ground truth in the second scene. (a) L-band CSAR. (b) Difference image by DBIFFPTD. (c) Detection result by DCFARFPTD. (d) Detection result by DBIFFPTD. (e) Ground truth.

with an efficient DBIF method, thus the time complexity of DBIFFPTD is much lower than that of CGANFPTD (105 s versus 8.2 h). Compared with DCFARFPTD which detects the target directly from the P-band LSAR image, the DBIF method in DBIFFPTD requires an additional time, thus the time complexity of DBIFFPTD is higher than that of DCFARFPTD (105 versus 45 s). Therefore, DBIFFPTD shows the best detection result among DCFARFPTD, CGANFPTD, and DBIFFPTD with moderate time complexity.

2) CSAR System: The observation scene in Figs. 17–20 is the urban street with various street lamps hidden under trees. Compared with the LSAR system, the CSAR system as a particular spotlight SAR system can achieve a high resolution [27], [28]. Besides, the CSAR image can fuse multiangle imaging results and that leads to anisotropic targets [29], [30], for example, partial covert targets presented in the CSAR image.

The detailed system parameters of this SAR system are listed in Table I. As we can see from Figs. 17(a) and 19(a) that several street lamps under trees can be weakly presented in the Ku-band CSAR image because several street lamps cannot be completely hidden in some azimuth angles by small trees. However, Figs. 17(b) and 19(b) could clearly present the street lamps because L-band radar wave can penetrate the foliage easily and the street lamp has the isotropic scattering. Figs. 17(c) and 19(c) show the DBIF-based fake Ku-band CSAR image which transfers similar texture information from the Ku-band CSAR image to the L-band CSAR image.

TABLE I

CSAR SYSTEM PARAMETERS

Parameter	Value
Carrier frequency f_c	Ku band, L band
Bandwidth	900M
The velocity of platform	50m/s
The height of platform	2000m
The radius of circular flightpath	2000m
The arc length of subapertures	$3^\circ \sim 10^\circ$

Considering that the imaging scene of the CSAR system is much smaller than that of the LSAR system and the co-registered dual-band CSAR training sample for CGAN is deficient, CGANFPTD could not be performed and selected as a comparison of the proposed DBIFFPTD. For the first scene in Fig. 18 and Table II, both DCFARFPTD and DBIFFPTD

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TABLE II
PERFORMANCE COMPARISON IN THE FIRST SCENE

	Precise targets	Missing targets	False alarms
DCFARFPTD [4]	18	3	15
DBIFFPTD	18	3	4
Ground Truth	22	0	0

TABLE III
PERFORMANCE COMPARISON IN THE SECOND SCENE

	Precise targets	Missed targets	False alarms
DCFARFPTD [4]	12	0	11
DBIFFPTD	11	1	6
Ground Truth	12	0	0

detect 18 covert targets marked by red frames and miss three covert targets marked by green frames but DBIFFPTD generates four false-alarm targets marked by blue frames which are much fewer than it of DCFARFPTD. For the second scene in Fig. 20 and Table III, DCFARFPTD could precisely detect all covert targets marked by red frames which is one more than the result of DBIFFPTD but DBIFFPTD generates six false-alarm targets marked by blue frames which are about half of it of DCFARFPTD. Balancing the number of precise targets, missing targets, and false-alarm targets, we consider DBIFFPTD to be superior to DCFARFPTD in detecting FP targets. Besides, the experimental result proves that DBIFFPTD can also steadily perform on the CSAR system.

VI. CONCLUSION

In this article, we are the first to take the DBIF method into the texture information transferring to get the fake high-band SAR image from the low-band SAR image, then the difference operator and the threshold segmentation are further utilized to extract changes between the high-band SAR image and the fake high-band SAR image. Based on the extracted change information, the DBIFFPTD method is developed to detect FP targets accurately and efficiently. Experimental data present that DBIFFPTD could well transfer the texture information of low-band SAR images to high-band SAR images and alleviate the false alarm which disturbs the detection result of the DCFAR-based FPTD (DCFARFPTD) method and the CGAN-based FPTD (CGANFPTD) method. Besides, DBIFFPTD consumes a moderate time complexity which would be relatively higher than it of DCFARFPTD and be massively lower than it of CGANFPTD.

REFERENCES

[1] I. Walterscheid, P. Berens, and M. Caris, "Potentials of multi-aspect and multi-frequency radar imaging illustrated by experimental results in Ka-and W-band," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6811–6822, Nov. 2018.

[2] M. Reza et al., "Multi-static multi-band synthetic aperture radar (SAR) constellation based on integrated photonic circuits," *Electronics*, vol. 11, no. 24, p. 4151, Dec. 2022.

[3] Y. Wang, X. Wang, C. Zang, W. Zhao, G. Cui, and S. Guo, "Multi-polarization features fusion detection of marine small targets based on LSTM," in *Proc. IEEE Int. Radar Conf. (RADAR)*, Nov. 2023, pp. 1–5.

[4] L. M. Novak, S. D. Halversen, G. Owirka, and M. Hiett, "Effects of polarization and resolution on SAR ATR," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 33, no. 1, pp. 102–116, Jan. 1997.

[5] D. He, T. Jin, and Y. Song, "A CGAN-based FOPEN target detection method for bi-frequency SAR," in *Proc. Int. Conf. Image, Video Process. Artif. Intell.*, 2020, pp. 1–6.

[6] Y. Zheng, X. Zhang, B. Hou, and G. Liu, "Using combined difference image and k-means clustering for SAR image change detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 11, no. 3, pp. 691–695, Mar. 2014.

[7] H. Noh, J. Ju, M. Seo, J. Park, and D.-G. Choi, "Unsupervised change detection based on image reconstruction loss," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 1351–1360.

[8] J. Huang, D. An, L. Chen, D. Feng, and Z. Zhou, "An NSST-based fusion method for airborne dual-frequency, high-spatial-resolution SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 4362–4370, 2023.

[9] C. Song et al., "A novel imaging-based target detection and parameter estimation scheme for airborne multichannel circular stripmap SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5203515.

[10] G. Easley, D. Labate, and W. Lim, *Sparse Directional Image Representations using the Discrete Shearlet Transform*. New York, NY, USA: Academic, 2008.

[11] J. Wu et al., "Multi-focus image fusion based on NSST and NLMF," *Inf. Technol. Netw. Secur.*, vol. 40, no. 5, pp. 39–44, Apr. 2021.

[12] A. L. Da Cunha, J. Zhou, and M. N. Do, "The nonsubsampled contourlet transform: Theory, design, and applications," *IEEE Trans. Image Process.*, vol. 15, no. 10, pp. 3089–3101, Oct. 2006.

[13] X. Wang, Y. Zhong, L. Zhang, and Y. Xu, "Spatial group sparsity regularized nonnegative matrix factorization for hyperspectral unmixing," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 11, pp. 6287–6304, Nov. 2017.

[14] D. Lee and and H. Seung, "Algorithms for non-negative matrix factorization," in *Proc. Neural Inf. Process. Syst. (NeurIPS)*. Cambridge, MA, USA: MIT Press, 2000, pp. 1–7.

[15] Z. Huang and J. He, "A novel image fusion algorithm based on NSCT and regional features," *J. Fuzhou Univ. (Natural Sci. Ed.)*, vol. 45, pp. 80–90, Jan. 2017.

[16] J. Zhang, M. Xing, G.-C. Sun, and Z. Wang, "Multiple statistics contributing to few-sample deep learning for subtle trace detection in high-resolution SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5210114.

[17] Y. Bai and J. Kang, "A global-to-local thresholding method for object enhancement in SAR images," in *Proc. 3rd China Int. SAR Symp. (CISS)*, Shanghai, China, Nov. 2022, pp. 1–4.

[18] J. Xue and D. Titterington, "T-tests, F-tests and Otsu's methods for image thresholding," *IEEE Trans. Image Process.*, vol. 20, no. 8, pp. 2392–2396, Aug. 2011.

[19] R. Colwell, *Manual of Remote Sensing*, 2nd ed. Oak Ridge, TN, USA: U.S. Department of Energy Office of Scientific and Technical Information, 1985.

[20] B. F. Fitriani, H. Wijanto, and A. D. Prasetyo, "Design of dual band microstrip antenna at L-band and S-band frequencies for synthetic aperture radar (SAR) sensors," *Jurnal Infotel*, vol. 10, no. 1, pp. 15–21, Feb. 2018.

[21] N. Zheng, Z. Yang, Z. Shi, H. Yang, Y. Sun, and F. Wang, "Airborne multidimensional SAR land cover dataset and fusion classification method," *Nat. Remote Sens. Bull.*, vol. 28, no. 9, pp. 2209–2222, Sep. 2024.

[22] Q. Lin and J. P. Allebach, "Combating speckle in SAR images: Vector filtering and sequential classification based on a multiplicative noise model," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 4, pp. 647–653, Jul. 1990.

[23] Y. Mou, Z. Wu, H. Zhao, and G. Wu. Allebach, "The terahertz scattering analysis of rough metallic and dielectric targets," *J. Radars*, vol. 7, no. 1, pp. 83–90, Jan. 2018.

[24] P. R. B. Da Silva, J. G. Vinholi, D. I. Alves, and R. Machado, "Enhancing change detection in ultra-wideband VHF SAR imagery: An entropy-based approach with median ground scene masking," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5208915.

[25] T. Hu, X. Zhang, Z. Sun, S. Ye, and P. Gao, "Contrasting effects of structural similarity and entropic metrics on band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[26] G. Gallardo i Peres, J. Dall, P. J. Mason, R. Ghail, and S. Hensley, "A generalized beta prime distribution as the ratio probability density function for change detection between two SAR intensity images with different number of looks," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5206414.

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[27] T. Watanabe, H. Yamada, Y. Tokutake, R. Sato, and Y. Yamaguchi, "Retrieval of polarimetric azimuthal angular characteristics via the application of target decomposition to spectral domain circular SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 5, pp. 2963–2982, May 2019.
[28] C. Yang, F. Zhao, Y. Deng, K. Liu, F. Zhao, and W. Wang, "Extended polar format algorithm and video-SAR image generation scheme for very high-resolution curvilinear spotlight SAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5215716.
[29] Y. Li, Q. Yin, Y. Lin, and W. Hong, "Anisotropy scattering detection from multiaspect signatures of circular polarimetric SAR," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 10, pp. 1575–1579, Oct. 2018.
[30] Y. Zhao, Y. Lin, W. Hong, and L. Yu, "Adaptive imaging of anisotropic target based on circular-SAR," *Electron. Lett.*, vol. 52, no. 16, pp. 1406–1407, Aug. 2016.



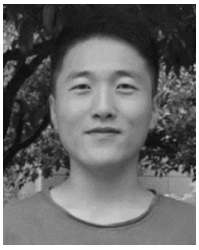
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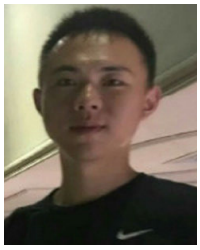
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